MoCap Tele OpenArm-Z



Designed For **Humanoid Robot** Teleoperation

ZeroLab Technologics Co.,Ltd

OpenArm & ROHand

MoCap Tele-OpenArm-Z Teleoperation System is a remote teleoperation platform designed for real-world humanoid robot manipulation tasks, combining **full-body motion capture** with an **embodied-intelligence robotic system**.

The system uses the **ZeroLab F2 Pro full-body motion capture device** as the human input interface to drive the **OpenArm humanoid robotic arm** equipped with the **ROHand dexterous hand**, enabling **low-latency, high-fidelity mapping** from human motion to robot action.

By capturing fine-grained movements of the operator’ s body, arms, and fingers through a wearable MoCap setup, the system translates natural human motions into real-time robotic actions in physical environments. It is well suited for **data collection, remote teleoperation, demonstration learning, and humanoid manipulation capability training**.

7DOF
Bimanual Arms

633mm
Arm Reach

6kg
Weight per Arm

6.0kg
Peak Payload per Arm

1kHz
CAN-FD Control

ROH-LiteS
dexterous hand

11 motion joints, 6 active ones
dexterous hand DoF





Core Value

- **Natural teleoperation**: The robot moves as the human moves, with minimal learning curve
- **High-precision data capture**: Outputs pose, quaternions, and finger-bending data to support imitation learning and reinforcement learning
- **Low-latency wireless operation**: System latency ⩽ 7.5 ms, enabling remote teleoperation across locations or cities and reliable performance in complex experimental and field environments
- **Complete “arm + hand” closed loop**: OpenArm provides safe, compliant humanoid arm motion, while ROHand delivers practical, high-performance dexterous manipulation
- **Open and extensible**: Open hardware and system architecture support research, enterprise customization, and large-scale deployment

Application Scenarios

- Humanoid robot teleoperation and demonstration
- Manipulation data collection (Manipulation Datasets)
- Embodied intelligence / Physical AI training
- Validation of contact-rich tasks (grasping, tool use, assembly)

F2 - Pro+ **Humanoid Robot** MoCap Suit

·Improved wearing experience

A streamlined and efficient wearability-focused design allows users to put on motion-capture equipment more quickly and comfortably, while enabling second-level data connection and calibration. This greatly reduces the time required for setup and use.

·longer wireless transmission range

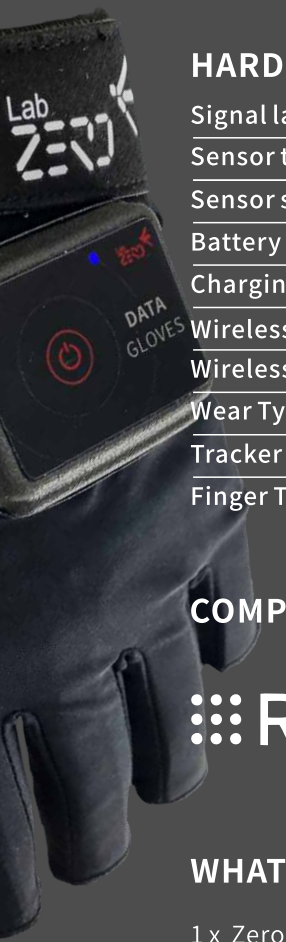
The upgraded F2 Pro+ supports a signal transmission range of up to 50 meters, allowing users to capture and transmit motion data in a larger and more flexible space. It also delivers stable performance even in complex signal environments.

·More DoF for the hand Tracking

The upgraded data gloves equipped on the F2 - Pro+ are specifically designed for 80% of the 6-DoF dexterous hands on the market, ensuring seamless compatibility with the vast majority of commercial robotic hands and delivering greater universality for humanoid robot dexterous-hand integration.

·Higher-precision motion data

The F2 Pro + full-body motion capture system delivers motion data accuracy almost on par with optical motion capture systems, while offering lower cost and greater operational flexibility. It combines high precision with low cost, making it ideal for large-scale deployment.



HARDWARE SPECS

Signal latency	⩽7.5ms(Determined by operating system limitations)
Sensor type	Euler angles, quaternions, finger bending resolution
Sensor sample rate	50Hz
Battery duration	Up to 10 hours
Charging time	0.5 hours
Wireless communication	Zero Lab 2.4G Proprietary protocol
Wireless range	Up to 50 meters
Wear Type	Strap-based
Tracker Quantity	15 + 1 Pair of Data Gloves
Finger Tracking	Zero Lab G3 Data Gloves



COMPATIBLE WITH

ROS · ubuntu · Windows 11 · UNREAL ENGINE · Unity

WHAT' S IN THE BOX

1 x ZeroLab G3 Data Gloves
1 x 1 to 2 Gloves Charging Cable
1 x Travel Case
1 x Power Plugs
15 x Zero Lab Motion Tracker
1 x Tracker Management Charger Unit
15 x Tracker Base
1 x Zero Lab Signal receiever
10 x Tracker Straps
1 x Zero Lab Motion Capture Master Software

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